Integral images of Springer–Borel maps

An introduction in the notation of Parts II and III

1. Introduction. The relation between representation rings and the cohomology of flag varieties has a classical model in the homomorphism from polynomial representations of GL_k to the integral cohomology of a Grassmannian. Under this map, irreducible representations indexed by partitions in the appropriate rectangle map to Schubert classes. Springer morphisms extend this construction to a representation-dependent setting for reductive groups.

Let G be a connected complex reductive group, B a Borel subgroup, $T \subset B$ a maximal torus, and $P \supset B$ a parabolic subgroup with Levi subgroup $L \supset T$. Write $\mathfrak{g}, \mathfrak{t}$ for the Lie algebras and W_L for the Weyl group of L. An irreducible representation $V(\lambda)$ with finite kernel determines the Springer morphism $\theta_\lambda : G \to \mathfrak{g}$, obtained by trace-orthogonal projection of its matrix realization onto $\mathfrak{g}$. Its pullback on T identifies $S(\mathfrak{t}^*)^{W_L}$, the invariant polynomial functions on $\mathfrak{t}$, with the subalgebra $\mathrm{Rep}^{\mathbb{C}}_{\lambda\text{-poly}}(L)$ of the complexified representation ring. The Borel characteristic map sends a weight to the first Chern class of its associated line bundle on G/B. Transporting its restriction to W_L-invariants through this identification gives

$$\xi^P_\lambda : \mathrm{Rep}^{\mathbb{C}}_{\lambda\text{-poly}}(L) \longrightarrow H^*(G/P, \mathbb{C}).$$

This is the construction of Kumar [K16]. For the defining representations of the classical groups we abbreviate λ-poly to poly.

Kumar–Rogers [KR20] studied the restrictions to the symplectic and odd orthogonal factors of the Levi subgroups. Their maps $\xi_{n,k}$ and $\bar{\xi}_{n,k}$ have targets $H^*(\mathrm{IG}(n-k, 2n), \mathbb{C})$ and $H^*(\mathrm{OG}(n-k, 2n+1), \mathbb{C})$, respectively. They determined the generator images and proved that the induced stable maps ξ_k and $\bar{\xi}_k$ are injective [KR20, Theorems 16 and 29]. Here stability means the graded inverse limit under the standard embeddings as $n \to \infty$, with k fixed. Kumar–Xie [KX24] obtained the even orthogonal counterpart $\xi^D_{n,k}$, with target $H^*(\mathrm{OG}(n-k, 2n), \mathbb{C})$, and proved stable injectivity [KX24, Theorem 3.8]. We retain the superscript D from their introduction to distinguish this map from the symplectic one.

We keep the parameters n, k of these papers: $1 \le k \le n-1$ in type B and $n \ge 4,\ 2 \le k \le n-1$ in type D. The variety $\mathrm{OG}(n-k, N)$ parametrizes isotropic $(n-k)$-planes for a nondegenerate symmetric form on $\mathbb{C}^N$. On the torus put $\bar{t}_j = (t_j - t_j^{-1})/2$. On the rank-$k$ orthogonal factor put $\bar{h}_i = (h_i - h_i^{-1})/2$, identified with $\bar{t}_{n-k+i}$ by the Levi inclusion, and write

$$e_i = e_i((\bar{h}_1)^2, \dots, (\bar{h}_k)^2), \qquad \bar{h}_{1,k} = \bar{h}_1 \cdots \bar{h}_k, \qquad \bar{h}^2_{1,k} = e_k.$$

Here $e_i(\,\cdot\,)$ is the i-th elementary symmetric polynomial; $\bar{h}_{1,k}$ occurs only in type D. For either orthogonal map the published generator formula is

$$e_i \longmapsto c_i^2 + 2\sum_{j=1}^{i} (-1)^j c_{i+j} c_{i-j}. \tag{1}$$

We use $c_i = c_i(Q^B)$ in type B and $c_i = c_i(Q)$ in type D, where the indicated bundle is the quotient of the trivial ambient bundle by the tautological isotropic subbundle; $c_0 = 1$ and $c_i = 0$ outside its rank. These are [KR20, Proposition 24] and [KX24, Theorem 3.6].

The integral forms used in the earlier injectivity arguments are

$$\mathrm{Rep}^{\mathbb{Z}}_{\mathrm{poly}}(\mathrm{SO}(2k+1)) = \mathbb{Z}[e_1,\dots,e_k],$$
$$\mathrm{Rep}^{\mathbb{Z}}_{\mathrm{poly}}(\mathrm{SO}(2k)) = \mathbb{Z}[e_1,\dots,e_k] \oplus \bar{h}_{1,k}\,\mathbb{Z}[e_1,\dots,e_k].$$

The second decomposition is additive; multiplication uses $\bar{h}_{1,k}^2 = e_k$. These are the *chosen normalized polynomial $\mathbb{Z}$-forms* of [KR20, KX24]. As in those papers, use cohomological degrees $\deg e_i = 4i$ and $\deg \bar{h}_{1,k} = 2k$. Equation (1) and the integral Euler image show that the restrictions $\bar{\xi}^{\mathbb{Z}}_{n,k}$ and $\xi^{D,\mathbb{Z}}_{n,k}$ take values in integral cohomology.

For $X = \mathrm{OG}(n-k, 2n+1)$ or $\mathrm{OG}(n-k, 2n)$, let $L_{n,k}(d)$ be the image of the degree-$2d$ part of the corresponding normalized form. We study the finite quotient

$$S_{n,k}(d)/L_{n,k}(d), \qquad S_{n,k}(d) = \big(L_{n,k}(d) \otimes_{\mathbb{Z}} \mathbb{Q}\big) \cap H^{2d}(X,\mathbb{Z}). \tag{2}$$

The group $S_{n,k}(d)$ is the *saturation* of the image. Because integral cohomology has a Schubert basis, the inclusion $L_{n,k}(d) \subseteq S_{n,k}(d)$ has positive Smith factors $s_1 \mid \cdots \mid s_r$: after integral basis changes, it is diagonal with these entries. Thus $S_{n,k}(d)/L_{n,k}(d) \cong \bigoplus_j \mathbb{Z}/s_j\mathbb{Z}$. We retain unit factors, recording the rank as well as the quotient. This is an index inside the selected rational image; that image can have smaller rank than the ambient cohomology.

Our first result determines all these factors in type B. Put

$$\ell = \min(k, n-k), \qquad A = \max(k, n-k), \qquad r = \lfloor n/2 \rfloor, \qquad c = \lceil n/2 \rceil.$$

For a partition $\gamma \subseteq A^\ell$, meaning $A \geq \gamma_1 \geq \cdots \geq \gamma_\ell \geq 0$, set

$$b_i = \gamma_i + \ell - i, \qquad u(\gamma) = \#\{i : b_i < r\}, \qquad E_{n,k}(\gamma) = \#\{i : b_i \geq c + u(\gamma)\}. \tag{3}$$

Zeros are included to make γ have exactly ℓ parts, and $|\gamma| = \sum_i \gamma_i$.

Theorem A (type B). *For $1 \leq k \leq n-1$, the nonzero Smith factors of $\bar{\xi}^{\mathbb{Z}}_{n,k}$ in cohomological degree $4m$ are the multiset*

$$\{2^{E_{n,k}(\gamma)} : \gamma \subseteq A^\ell,\ |\gamma| = m\}. \tag{4}$$

The image is zero in cohomological degrees not divisible by four. In particular every factor is a power of two, with exponent at most ℓ.

The statistic labels multiplicities; it does not assert that the individual Schur polynomials already give a Smith basis. For the type-D statement, package these multiplicities as

$$\mathcal{B}_{n,k,e}(z) = \sum_{\substack{\gamma \subseteq A^\ell \\ E_{n,k}(\gamma) = e}} z^{2|\gamma|}. \tag{5}$$

Thus z^d records cohomological degree $2d$, and $e \geq 0$ records the factor 2^e. Set $\mathcal{B}_{k,k,0} = 1$ and $\mathcal{B}_{k,k,e} = 0$ for $e > 0$; this is the point boundary needed below. Parameters in (3) are always recomputed for the displayed pair n, k.

The distinction between normalized and literal character lattices matters. The chosen $\mathrm{Rep}^{\mathbb{Z}}_{\mathrm{poly}}$ need not equal the intersection with the ordinary integral representation ring. The alternative character lattice generated by $2^{2i}e_i$ and, in type D, $2^k\bar{h}_{1,k}$, has degree-$2d$ image $2^d L_{n,k}(d)$; each Smith factor is therefore multiplied by 2^d. The theorems concern the normalized forms specified above.

Let $M_D(n,k;e,d)$ be the multiplicity of 2^e among the nonzero Smith factors of $\xi_{n,k}^{D,\mathbb{Z}}$ in $H^{2d}(\mathrm{OG}(n-k,2n),\mathbb{Z})$. Put $\mathcal{I}_{n,k}(z)=0$ for odd k. For $k=2h$, define

$$\mathcal{I}_{n,2h}(z)=z^{\,2h(\lfloor (n-2h)/2\rfloor+1)}\begin{bmatrix} h+\lceil (n-2h)/2\rceil-1 \\ h \end{bmatrix}_{z^2}. \tag{6}$$

Here the Gaussian polynomial $\begin{bmatrix} v \\ u \end{bmatrix}_q=\sum_{\mu\subseteq(v-u)^u} q^{|\mu|}$ counts partitions in a rectangle.

Theorem B (type D). *For $n\geq 4$ and $2\leq k\leq n-1$, every nonzero Smith factor of $\xi_{n,k}^{D,\mathbb{Z}}$ is a power of two, and for every $e\geq 0$,*

$$\boxed{\sum_{d\geq 0} M_D(n,k;e,d)z^d=\mathcal{B}_{n,k,e}(z)+z^k\mathcal{B}_{n-1,k,e}(z)+(\delta_{e,1}-\delta_{e,0})\mathcal{I}_{n,k}(z).} \tag{7}$$

Here δ is the Kronecker delta.

The first two terms are the factors of the square-generated image and the image obtained by multiplication by $\bar{h}_{1,k}$, respectively. When k is odd they occupy different parities of d. For even k, the sum of their separate saturations may have finite index in the saturation of their combined image; we call this *integral gluing*. The polynomial $\mathcal{I}_{n,k}$ counts every gluing direction. Each such direction replaces a pair of unit factors $(1,1)$ by $(1,2)$, with no further correction among the higher factors. For example, $n=4,k=2$ in cohomological degree 8 gives the full type-D list $(1,2,2)$, and hence quotient $(\mathbb{Z}/2\mathbb{Z})^2$.

The proofs refine the earlier generator calculations by integral incidence and lattice arguments. In type B, Schur-product filtrations are shown to be primitive, meaning that their quotients in the ambient free abelian lattice are torsion free. A unit triangular minor and a matching geometric rank bound determine these filtrations and yield (4). In type D, relative orthogonal incidence bundles, the integral Euler relation, and parity retractions establish the comparison of the square and Euler sources with type B. An index comparison then proves that the unit gluing exhausts all saturated gluing, giving (7). The integral Schubert calculus is published input; the finite-rank image classification is the additional deduction developed in the companion proofs [N]. Exact calculations corroborate the formulas in finite cases. Together, the results determine the degreewise integral divisibility of the selected Springer–Borel images, refining the explicit images and stable injectivity of Parts II and III.

References

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[KR20] S. Kumar and S. Rogers, *Representation ring of Levi subgroups versus cohomology ring of flag varieties II*, J. Algebra **556** (2020), 340–362. Author text.

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[N] *Integral Springer–Borel image notes*, unpublished companion proofs, audited 19 October 2026. Type B; type D; audit. The deductions are internally checked; external expert review and originality remain unestablished.